

Sequential reference preservation for label-efficient set-valued prediction

Xin Feng^{a,1}, Decong Yang^{a,2,*}

^a*Xinjiang University, Youhao Campus, 499 Xibei Road, Shayibake District, Urumqi, 830091, Xinjiang, China*

Abstract

Label-efficient calibration requires a criterion for when acquired labels suffice to preserve a reference predictor's decisions. We formulate sequential reference preservation with a fixed score function and prescribed set-size tolerance, and characterize an observable certificate for stopping acquisition. For any quantile interval, the mean cardinality difference between its endpoint prediction sets equals the smallest worst-case enlargement achievable by a predictor required to contain every set generated within that interval. The upper-endpoint predictor attains this bound. This characterization makes unresolved candidate labels the quantity that determines whether the current interval certifies the required precision. Joint rank calibration (JRC) implements the procedure with hypergeometric rank bounds calibrated over declared checkpoints. Its recursion jointly tracks ranks below the reference rank and inclusion of the reference-ranked item. Under uniform sampling without replacement, the returned sets contain the complete-reference sets for every input and satisfy the monitoring-batch enlargement tolerance on one confidence event. Evaluation covers 354 classifier cases on seven wearable datasets and 45 additional Heterogeneity Human Activity Recognition (HHAR) cases. At confidence error 0.05 and tolerance 0.5, JRC saves 23.67% of labels on the University of Southern California Human Activity Dataset (USC-HAD), compared with 17.91% for Bonferroni hypergeometric bounds, 11.74% for a uniform-prior confidence sequence and 15.33% for its strongest reported prior variant. HHAR savings are 34.03%, compared with 25.13% for the fixed Beta(9, 1) comparator.

Keywords: Set-valued prediction, Reference preservation, Sequential calibration, Finite-population inference, Label acquisition, Conformal prediction

1. Introduction

A calibrated prediction set is useful when it retains plausible outcomes without burdening a decision maker with too many alternatives. Conformal prediction provides a statistical framework for these sets [1, 2], connected to broader representations of uncertainty [3]. Acquiring calibration labels creates a second decision: when has labeling resolved enough uncertainty to produce the required sets? We study this question relative to a fixed reference pool. The aim is to preserve the prediction sets that complete labeling would produce, with a prescribed average allowance for additional candidate labels.

This objective gives label acquisition a different target from related approaches. Ilyas et al. [4] optimize a combined labeling and prediction-set cost under an independent and identically distributed (i.i.d.) model; the target here is containment of a finite reference predictor within a

*Corresponding author

Email address: 20232501525@stu.xju.edu.cn (Decong Yang)

¹ORCID: <https://orcid.org/0009-0008-3242-7961>

²ORCID: <https://orcid.org/0009-0009-9295-5149>

set-size tolerance. Confidence sequences [5–7] supply sequentially valid intervals; reference preservation uses those intervals to certify the precision of the resulting sets. Active risk-controlling prediction sets [8] select labels to support a prediction-risk guarantee; the present objective controls the loss and enlargement of specified reference decisions. The distinction is observable in the output: every reference candidate label is retained on the confidence event, and the mean number of additional candidates is bounded on the monitoring batch.

The central quantity is the unresolved candidate-label mass inside the current quantile interval. Numerical interval width alone does not determine this mass: a narrow interval can cross many candidate scores, while a wider interval can leave all set memberships unchanged. We characterize the endpoint set-size difference as a tight precision certificate. Among outputs required to contain the sets generated by every threshold in the interval, the upper-endpoint predictor is pointwise smallest. Its worst-case mean enlargement is exactly the difference between the upper- and lower-endpoint set sizes. Consequently, the set-size tolerance can be certified from that interval precisely when this difference is small enough. This provides a decision-based interpretation of both the returned predictor and the stopping criterion.

Joint rank calibration (JRC) supplies a finite-pool implementation of this principle. It combines hypergeometric rank inference with exact group-sequential calibration on a declared query schedule [9, 10]. One state component counts sampled ranks below the reference rank; a binary component records inclusion of the reference-ranked item. The lower boundary depends on their sum and the upper boundary on the count, so their joint propagation evaluates crossings along the same query path. At each checkpoint, unlabeled candidate scores convert the quantile bracket into its precision certificate. Querying stops once that certificate meets the tolerance, and the upper-endpoint sets are returned.

The contributions are:

1. A reference-preservation acquisition objective combining retention of the complete-reference sets with a directly interpretable budget for additional candidate labels.
2. A characterization of the smallest worst-case enlargement under interval uncertainty, together with a sequential procedure that certifies this tolerance at its stopping time.
3. A finite-checkpoint rank-calibration implementation and an evaluation on seven wearable datasets with 354 classifier cases, plus 45 additional HHAR cases, relating label savings to the reference sets, boundary construction and precision budget.

The framework separates the probability budget for reference loss from the cardinality budget for added labels. The latter is measured in decision space and is invariant to strictly increasing score transformations. Wearable recognition provides an application in which user heterogeneity and probability calibration affect prediction-set utility [11, 12]. The paired experiments evaluate JRC and the stated hypergeometric and confidence-sequence comparators using the same pools, query orders and precision tolerances.

2. Related work

2.1. Set-valued prediction and uncertainty representation

Prediction sets make the trade-off between error control and output precision explicit. Least-ambiguous set-valued classification studies small sets subject to error constraints [13], while conformal methods connect set construction to calibration ranks [1, 2, 14]. Cella and Martin relate conformal prediction to consonant plausibility measures and examine validity for different prediction tasks [3]. The object considered here is the enclosure of a fixed reference predictor as its calibration labels become available. Analyses of expected conformal set size quantify efficiency

over calibration and prediction randomness [15]. Our precision certificate evaluates enlargement on the fixed monitoring batch for every threshold in the current interval.

User and distribution heterogeneity affect the calibration properties of that reference. Hierarchical prediction and methods beyond exchangeability provide tools for such settings [16, 17]. Our acquisition target conditions on the chosen finite pool and refines its quantile through randomized label queries. This separates uncertainty about the reference threshold from the ambiguity represented by the reference sets themselves.

2.2. Sequential inference and label acquisition

Ilyas et al. [4] study a stream of n independent and identically distributed (i.i.d.) examples and the cost

$$C(m) = m + (n - m) \mathbb{E}[|T_m(X)|]/K. \quad (1)$$

Their procedure estimates normalized set size from candidate-label scores on unlabeled inputs and stops using a lower confidence bound on the cost increment. The regret analysis uses a monotone score-density-ratio assumption. The original-submission manuscript also describes a finite-sample stabilization for the experiments. These ideas motivate stopping decisions based on the observable size of prediction sets.

JRC targets containment of the prediction sets defined by a fixed reference pool. Its confidence statement is conditional on that pool and the uniform query order, with an additional size constraint on the monitoring batch. Table 1 summarizes the statistical targets of the related approaches. The Ilyas comparison follows the original-submission version and distinguishes its theoretical stopping rule from the experimental stabilization.

Training-conditional validity has a longer history [18]. Liang et al. study conformal inference with early stopping of neural-network training and reuse of hold-out data [19]. Xu et al. develop active, anytime-valid risk-controlling sets using a labeling policy and risk-predictor assistance [8]. The August 2026 preprint by Skalse et al. [20] studies a game with prediction and label-query actions on non-independent streams. Together, these studies address limited labeling resources through several population-risk and sequential-decision objectives.

Table 1: Related statistical targets and acquisition decisions

Procedure	Sampling target	Stopping or acquisition decision	Guarantee considered
Ilyas et al. [4]	I.i.d. data-generating distribution	Cost-increment rule; empirical set size estimated from unlabeled inputs	Population coverage; asymptotic cost regret under a density-ratio condition
Howard and Ramdas [6]	Population quantiles from an i.i.d. stream	Any stopping time supported by the confidence sequence	Time-uniform confidence intervals for quantiles and distribution functions
Waudby-Smith and Ramdas [7]	Fixed population sampled without replacement	User-selected stopping applied to a confidence sequence	Time-uniform confidence sets for finite-population parameters
JRC	Corrected order statistic of a fixed finite reference pool	Endpoint set-size difference below a batch-specific tolerance	Reference-set containment for all inputs; average inflation control on the monitoring batch

All methods are compared under their stated sampling target and stopping contract.

Finite-population sampling and hypergeometric inversion provide rank-inference tools [9]. Confidence sequences [5–7] control sequential inspection, while exact group-sequential designs calibrate crossing probabilities on a chosen schedule [10]. The present procedure uses these principles to bound a fixed reference quantile and connects the bounds to an observable prediction-set stopping criterion.

2.3. Wearable recognition as an evaluation setting

Recent HAR frameworks improve representation learning and adaptation. STF-AL combines temporal, spatial and frequency representations with adversarial domain adaptation [21]. CCT-HAR uses cross-modal sample mining and temporal alignment [22], and TSDFS studies attention-based feature selection for multiple sensors [23]. These methods motivate controlled evaluation of each component in a recognition pipeline. Our comparisons hold the trained classifier fixed to isolate the effect of calibration-label acquisition.

Uncertainty estimation is also important in HAR. Roy et al. study confidence-calibrated recognition [24], and hyperdimensional computing combined with conformal prediction supports uncertainty-aware inference on microcontrollers [25]. MiniRocket [26] and InceptionTime [27] provide established model families with contrasting representations. We use them to examine how acquisition behavior changes with the underlying score distribution.

3. Finite-pool reference emulation

Figure 1 shows the frozen-model and query workflow. The running extrema use the visited checkpoints, and both sums range over the monitoring batch. The checkpoint set in panel (b) denotes $\mathcal{T}$, whereas T is the stopping query count. Algorithm 1 specifies the acquisition procedure.

3.1. Target, information and cost

Let a trained classifier be fixed before calibration acquisition. For a nonconformity score $s(x, y)$, its thresholded prediction set is

$$C(x; q) = \{y \in \mathcal{Y} : s(x, y) \leq q\}. \quad (2)$$

The label set $\mathcal{Y}$ is finite, with K classes. We fix a reference pool $\mathcal{P} = \{x_i\}_{i=1}^N$ before requesting its labels. The unknown true-label scores $S_i = s(x_i, y_i)$ are regarded as fixed values. Define

$$k = \lceil (N + 1)(1 - \alpha) \rceil, \quad q_{\text{ref}} = S_{(k)}, \quad (3)$$

with $q_{\text{ref}} = +\infty$ if $k > N$. The predictor $C(x; q_{\text{ref}})$ is the complete reference. Sequential acquisition aims to preserve these sets while using a subset of the pool labels. Population coverage is determined by the reference and its calibration assumptions; the probability statement below controls uncertainty arising from label acquisition.

We condition on all pool values, the trained classifier and a batch $\mathcal{X}_\star$ of M unlabeled monitoring inputs. The probability statements concern a uniform permutation of the N pool indices, selected independently of their unrevealed labels. This design accommodates dependencies among people, repeated windows and trials because inference conditions on the complete finite pool.

Candidate-label scores on $\mathcal{X}_\star$ are available before acquisition. They turn a quantile interval into an observable enclosure of the reference prediction sets. The stopping criterion controls the mean width of this enclosure on the monitoring batch.

The primary cost counts T queried window labels. With m reference participants and setup cost c per participant, modeled effort is $T + cm$, compared with $N + cm$ for the complete reference. Both policies include all participant setup costs; signal collection, segmentation and pool construction precede acquisition.

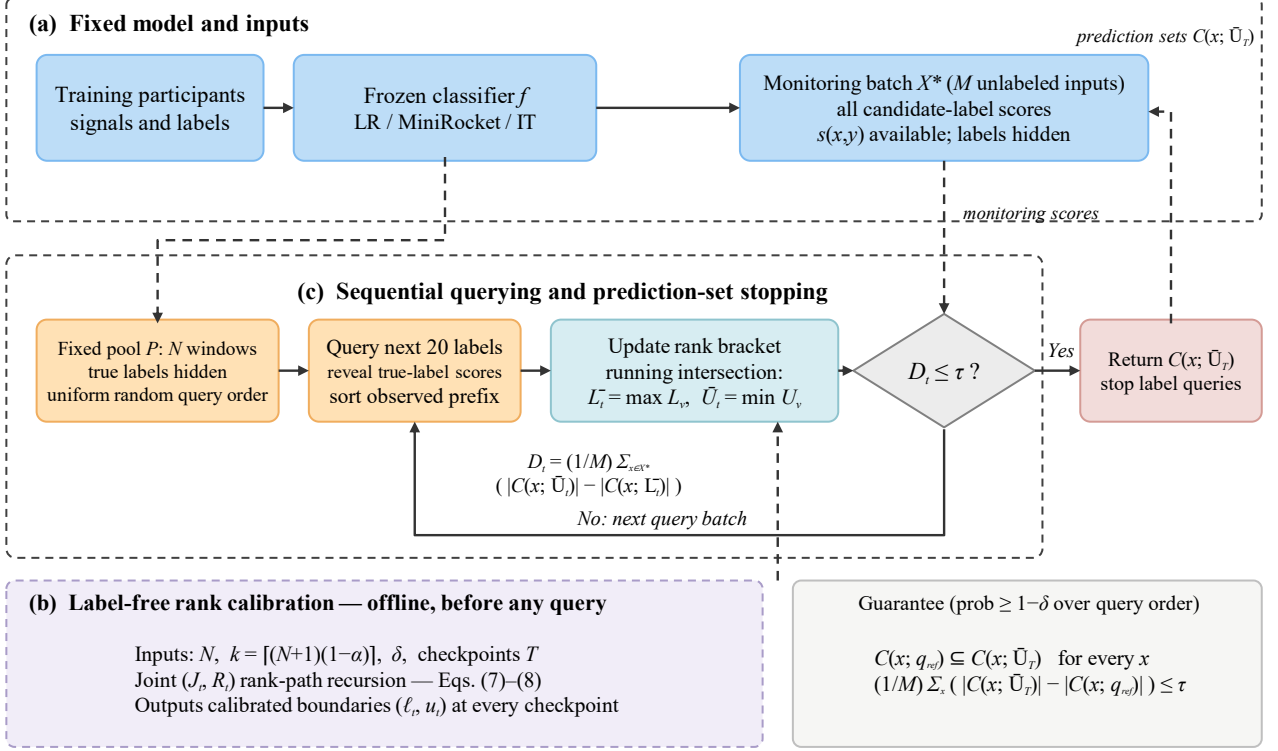


Figure 1: JRC framework. (a) A frozen classifier provides candidate-label scores. (b) Joint rank calibration constructs checkpoint boundaries before querying. (c) Labels are queried uniformly without replacement to update the quantile bracket; querying stops when the endpoint sets meet the monitoring-batch size tolerance, and the upper-endpoint sets are returned

3.2. Finite-population rank brackets

Choose checkpoints $\mathcal{T} \subseteq \{1, \dots, N\}$ with $N \in \mathcal{T}$. If scores tie, order the complete population by the pair (S_i, i) , using the fixed pool index to break ties. The numerical inequalities below remain valid when these pairs are projected onto their score values.

Let J_t count the queried items whose full-population ranks are strictly below k , and let R_t indicate whether the item of rank k has appeared. Then $J_t + R_t$ counts sampled items with rank at most k . Their separate marginal distributions are

$$\begin{aligned} J_t &\sim \text{Hypergeom}(N, k-1, t), \\ J_t + R_t &\sim \text{Hypergeom}(N, k, t). \end{aligned} \quad (4)$$

Write $Q_p(N, a, t)$ for the smallest integer whose hypergeometric cumulative probability is at least p . At a common tail parameter η , use sample-rank boundaries

$$\ell_t = Q_\eta(N, k, t), \quad u_t = Q_{1-\eta}(N, k-1, t) + 1. \quad (5)$$

For the ordered observed scores $S_{(j)}^{[t]}$, set

$$L_t = S_{(\ell_t)}^{[t]}, \quad U_t = S_{(u_t)}^{[t]}. \quad (6)$$

Here ℓ_t is the lower sample-rank index defined in Eq. (5). The conventions are $S_{(0)}^{[t]} = -\infty$ and $S_{(t+1)}^{[t]} = +\infty$. At the census, both bounds are the known $S_{(k)}$. When $k > N$, both are $+\infty$ and the reference already contains every label.

Algorithm 1 Joint rank calibration for a fixed reference pool

Require: Frozen score function; fixed pool of N windows; uniform query permutation; monitoring scores; α, δ, τ ; checkpoints $\mathcal{T}$

- 1: Set $k = \lceil (N+1)(1-\alpha) \rceil$ and calibrate rank boundaries (ℓ_t, u_t) by the joint recursion
Joint recursion: Section 3.3, Eqs. (7)–(8).
 - 2: Initialize observed scores $\mathcal{S} \leftarrow \emptyset$, $L \leftarrow -\infty$, $U \leftarrow +\infty$
 - 3: **for** each checkpoint $t \in \mathcal{T}$ **do**
 - 4: Query only the new labels needed to reach t observations and add their true-label scores to $\mathcal{S}$
 - 5: Obtain the current rank bracket (L_t, U_t) from the ordered observed scores
 - 6: Form $(L', U') \leftarrow (\max\{L, L_t\}, \min\{U, U_t\})$
 - 7: **if** $L' > U'$ **then**
 - 8: Record an empty-intersection event and use $(L', U') \leftarrow (L_t, U_t)$
 - 9: **end if**
 - 10: Set $(L, U) \leftarrow (L', U')$ and calculate D_t from the monitoring scores using Eq. (9)
 - 11: **if** $D_t \leq \tau$ **then**
 - 12: **return** queried count t and predictor $C(x; U)$
 - 13: **end if**
 - 14: **end for**
-

On a path satisfying $J_t + R_t \geq \ell_t$ and $J_t < u_t$, at least ℓ_t observations have full rank at most k , while fewer than u_t observations have rank below k . Consequently, $L_t \leq q_{\text{ref}} \leq U_t$. A standard simultaneous-coverage construction uses $\eta = \delta/[2(|\mathcal{T}| - 1)]$ and a union bound over the non-census checkpoints. The resulting Bonferroni hypergeometric (HG) brackets form the first baseline.

3.3. Joint calibration of the checkpoint boundaries

We calibrate the probability of crossing either rank boundary over the full checkpoint sequence. Conditional on state $(J_t, R_t) = (j, r)$, the next observation has rank below k , equal to k , or above k , with probabilities

$$\frac{k-1-j}{N-t}, \quad \frac{1-r}{N-t}, \quad \frac{N-k-t+j+r}{N-t}, \quad (7)$$

respectively. They lead to states $(j+1, r)$, $(j, 1)$ and (j, r) . Impossible transitions have probability zero. Start with mass one at $(0, 0)$; propagate these probabilities and absorb mass at any checkpoint for which

$$j+r < \ell_t \quad \text{or} \quad j \geq u_t. \quad (8)$$

The absorbed mass $A(\eta)$ is the exact probability of at least one rank-boundary crossing under the finite query design. There are at most $2k$ states at each time. Direct propagation costs $O(Nk)$ operations and $O(k)$ working memory for a specified boundary family. It is performed before label queries and reused across pools with the same N , k , confidence level and checkpoints.

Increasing η tightens the rank bracket and increases the crossing probability $A(\eta)$. We use 24 bisection steps and retain a feasible value satisfying $A(\eta) \leq \delta - 10^{-10}$. This implements exact group-sequential calibration [10] for the specified reference rank. The resulting tables are reusable across pools with the same finite query design.

3.4. Stopping in prediction-set space

Intersect valid past brackets by setting $\bar{L}_t = \max_{v \leq t} L_v$ and $\bar{U}_t = \min_{v \leq t} U_v$, where $v \in \mathcal{T}$. On the simultaneous coverage event this intersection contains q_{ref} . An empty intersection signals

that event has failed; the implementation records it and falls back to the current bracket. At the census it returns the exact quantile regardless of previous events.

The stopping statistic is observable without monitoring labels:

$$D_t = \frac{1}{M} \sum_{x \in \mathcal{X}_*} [|C(x; \bar{U}_t)| - |C(x; \bar{L}_t)|]. \quad (9)$$

For tolerance $\tau \geq 0$, querying stops at the first checkpoint with $D_t \leq \tau$ and returns $C(x; \bar{U}_T)$. The census guarantees termination. The tolerance counts additional candidate labels per monitoring input, so its interpretation is preserved by strictly increasing transformations of the score.

Proposition 1 (reference preservation). For a fixed classifier, finite reference pool, monitoring batch and declared checkpoints, suppose the joint recursion yields $A(\eta) \leq \delta$. Under uniform sampling without replacement, with probability at least $1 - \delta$, the returned predictor satisfies simultaneously

$$C(x; q_{\text{ref}}) \subseteq C(x; \bar{U}_T), \quad (10)$$

for every input x , and

$$\frac{1}{M} \sum_{x \in \mathcal{X}_*} [|C(x; \bar{U}_T)| - |C(x; q_{\text{ref}})|] \leq \tau. \quad (11)$$

No independence assumption on the fixed score values is required.

Proof. The recursion enumerates the law of the uniformly sampled rank path. Except on an event of probability $A(\eta)$, neither inequality in Eq. (8) holds at any checkpoint. The rank argument above gives $L_t \leq q_{\text{ref}} \leq U_t$ simultaneously. Running intersections therefore contain q_{ref} and are nonempty. Since the sets in Eq. (2) are nested in q , the upper endpoint gives Eq. (10) for every input. The reference set lies between the sets at the lower and upper endpoints, so its extra-label difference is bounded pointwise by the summand in Eq. (9). At the data-dependent stopping time, the average is at most τ . The confidence event covers every checkpoint and hence also this stopping time. $\square$

On the confidence event, empirical coverage is at least that of the reference on any fixed labeled test collection. The confidence level applies to one campaign under the declared query design.

Strictly increasing score transformations preserve the rank ordering and every thresholded set membership, so they leave the query count and returned sets unchanged. This invariance provides an implementation check.

A tight certificate under interval uncertainty. The stopping statistic also has a deterministic decision interpretation. For a nonempty interval $I = [L, U]$, let $\mathcal{A}(I)$ contain the set-valued predictors A satisfying $C(x; q) \subseteq A(x)$ for every input x and every $q \in I$. This is the information model in which the reference threshold can be any value in the certified interval. Write $D(L, U)$ for the endpoint difference in Eq. (9) with these bounds.

Proposition 2 (smallest worst-case enlargement). For the finite label space and fixed monitoring batch,

$$\inf_{A \in \mathcal{A}(I)} \sup_{q \in I} \frac{1}{M} \sum_{x \in \mathcal{X}_*} [|A(x)| - |C(x; q)|] = D(L, U).$$

The predictor $A(x) = C(x; U)$ attains the infimum and is pointwise smallest in $\mathcal{A}(I)$.

Proof. The containment condition at $q = U$ requires $A(x) \supseteq C(x; U)$ for every input. The admissible threshold $q = L$ therefore gives a mean enlargement of at least $D(L, U)$ for every such output. For $A(x) = C(x; U)$, nesting makes this difference largest at $q = L$, where it equals $D(L, U)$. This output attains the bound. $\square$

Thus $D(L, U) \leq \tau$ is necessary and sufficient for certifying the tolerance using this interval information and uniform containment requirement. Its value is the observed candidate-score mass between the endpoints:

$$D(L, U) = \frac{1}{M} \sum_{x \in \mathcal{X}_*} \sum_{y \in \mathcal{Y}} \mathbf{1}\{L < s(x, y) \leq U\}.$$

For example, under the binary least ambiguous set-valued classifier (LAC) score [13] $s(x, y) = 1 - \hat{p}_y(x)$, an interval $[0.49, 0.51]$ gives endpoint difference zero for class probabilities $(0.9, 0.1)$ and two for $(0.5, 0.5)$. The same numerical width therefore represents different amounts of unresolved prediction-set membership. The characterization explains how the current bracket certifies set precision; the statistical guarantee for the data-dependent query count is supplied by Proposition 1.

3.5. Implementation and comparison methods

Sorted monitoring scores allow Eq. (9) to be computed using two binary searches. Each checkpoint sorts the observed label scores. This direct implementation is adequate for the pool sizes studied; dynamic order-statistic structures could reduce repeated sorting for larger pools. Precomputed boundary tables are separate from per-pool query execution.

The interval methods share the reference pool, query order, upper-endpoint output and set-inflation tolerance. Their rank bounds satisfy the stated confidence requirement over their respective query schedules. This common acquisition procedure provides the basis for comparing label cost.

The second quantile-interval baseline inverts the prior-posterior ratio confidence sequence (CS) of Waudby-Smith and Ramdas [7]. A uniform prior on the number of successes in the population induces the ordered-prefix mixture likelihood $B(j + 1, t - j + 1)$. For a hypothesized population count a , divide this mixture likelihood by

$$\mathcal{L}_t(a, j) = \frac{(a)_j (N - a)_{t-j}}{(N)_t}, \quad (12)$$

where $(a)_j$ denotes a falling factorial. Invert the resulting e-process separately at counts k and $k - 1$, allotting each error $\delta/2$. The minimum accepted count at k gives the lower sample rank; one plus the maximum accepted count at $k - 1$ gives the upper rank. We use the same running intersection, checkpoints and prediction-set stopping tolerance as JRC. JRC is calibrated on the declared finite schedule, whereas the CS comparator is valid at every sampling time; this difference is part of the reported comparison.

We additionally compare ordinary conformal quantiles from fixed 25%, 50% and 75% prefixes, and an upper hypergeometric bound at a single fixed 50% look. The former have no reference-containment guarantee. The latter protects the reference at that fixed look but does not adapt label expenditure to a prescribed set-size tolerance. Diagnostic ablations omit time correction, omit running intersection, return a midpoint at the proposed stopping time, or use sparse checkpoints. Their different guarantees are kept explicit. The one-look HG comparator queries $\lceil N/2 \rceil$ labels and returns the upper endpoint of the same two-sided hypergeometric interval, with error 0.025 in each tail. Thus, “50% look” identifies the query fraction. This comparator uses a more conservative upper endpoint than a one-sided bound assigning error 0.05 to the upper tail.

4. Experimental design

4.1. Data and development history

The main evaluation includes six development datasets: UCI HAR [28], DSADS [29], MHEALTH [30], WISDM [31], PAMAP2 [32] and REALDISP [33]. They contain 122 eligible

participant identifiers and 54,482 retained windows. These datasets were examined during acquisition-policy development. Earlier risk-based and randomized-allocation rules showed poor tolerance transfer on PAMAP2 and REALDISP, motivating the fixed-reference target. Their protocols and aggregate negative results are retained in Supplementary material 1.

USC-HAD [34] provides the main confirmation benchmark. The method, confidence level, primary tolerance, query schedule, preprocessing and splits were locally locked on 12 September 2026 before model predictions were available. All 14 participants meet the requirement of at least 120 retained windows, yielding 5,353 windows and 12 classes. The main seven-dataset study therefore contains 136 dataset-specific participant identifiers and 59,835 windows.

The selected input is three acceleration axes resampled to 128 time points. UCI HAR uses the release’s 2.56-second total-acceleration windows, retaining alternate windows in the released subject order. DSADS uses five-second torso-acceleration segments. MHEALTH uses five-second chest windows; WISDM uses five-second smartphone-acceleration windows from its 18-activity release. PAMAP2 uses protocol chest acceleration, and REALDISP uses ideal-placement BACK acceleration. The original REALDISP manual places BACK third in the log, unlike the fifth position listed in the repository webpage summary; we follow the original manual. The retained sample counts and preprocessing provenance are included in Table 2 and the supplement. Table 2 records the window durations and sensor placements.

For USC-HAD, each original MAT trial remains separate. We take nonoverlapping 500-sample windows at the documented 100 Hz, retain acceleration columns 1–3, reject nonfinite windows and incomplete tails, and use anti-aliased polyphase resampling to 128 points. The acquisition study starts from the supplied trial segmentation. The filename convention supplies activity IDs. An independent schema audit finds one conflict: Subject14/a3t2.mat is activity 3 by filename but activity 2 (walking-left) in its internal metadata. Main labels follow the locked filename rule. A separate sensitivity excludes that trial, refitting classifiers whenever it occurs in training. Another file lacks the optional numeric activity field but has a matching activity-name string; it is retained. No exact duplicate signal arrays are found among the 840 trials. The main analysis and the exclusion sensitivity are both reported.

Table 2: Public datasets and retained samples

Dataset	People	Windows	Classes	Selected signal	Seconds
UCI HAR	30	5,155	6	Waist total acceleration	2.56
DSADS	8	9,120	19	Torso acceleration	5
MHEALTH	10	1,335	12	Chest acceleration	5
WISDM	51	32,653	18	Phone acceleration	5
PAMAP2	8	3,775	12	Chest acceleration	5
REALDISP	15	2,444	33	BACK acceleration	5
USC-HAD	14	5,353	12	Front right hip acceleration	5

The first six datasets are developmental; USC-HAD is the locked confirmation. Counts follow the fixed preprocessing before schema sensitivity.

4.2. Frozen classifiers and subject partitions

Every acquisition method uses identical probability vectors within a classifier case. Logistic regression (LR) uses 35 standardized statistical features, regularization $C = 1$ and a maximum of 2,000 iterations. MiniRocket (MR) uses 2,016 kernels followed by a standardized logistic-regression head. The InceptionTime-style model (IT) uses six modules, 32 filters per branch, kernel lengths 39, 19 and 9, residual connections every third module, global pooling and a classification head. It is trained for 40 epochs with Adam, learning rate 10^{-3} and batches of 128, without

a test-selected checkpoint. The statistical features, normalization and model code are shared with the preceding frozen model study. Scikit-learn and SciPy support the classical pipeline [35, 36].

We reuse 249 cases from 83 outer partitions on the six development datasets. Each case has disjoint training, calibration and test participants. The acquisition procedure uses the frozen final-classifier predictions. USC-HAD has five fixed random seeds and seven test folds per seed: two test participants, eight training participants and four calibration participants in each fold. This gives 35 partitions and 105 new final-classifier fits. The 354 model cases reuse the 83 development and 35 confirmation partitions; repeated results are aggregated within participants.

The least ambiguous set-valued classifier (LAC) score [13] is $s(x, y) = 1 - \hat{p}_y(x)$. The secondary score uses adaptive prediction sets (APS) [14] in a deterministic inclusive form: sort class probabilities in descending order, with stable class-index ties, and assign each class its cumulative probability. There is no randomization within the APS score and no forced top-label insertion. Empty and full sets remain in the calculations. The same scoring convention is used for the reference and every acquisition comparator, and $\alpha = 0.1$ throughout.

4.3. Reference pools and query experiments

We construct each reference pool from m calibration participants. The requested ceilings are $B \in \{120, 240, 480, 960\}$, and the actual pool size is $N = \min(B, 120m)$. Participant quotas differ by at most one; a random participant ordering allocates any remainder. Each quota is sampled uniformly without using its labels, after which the complete selected pool is independently permuted. When several ceilings produce the same pool size, the evaluation retains the declared configurations and reports their actual N .

The primary settings are confidence error $\delta = 0.05$, mean set-inflation tolerance $\tau = 0.5$, and checkpoints every 20 queried labels plus the census. The tolerance is measured in additional candidate labels per monitoring input. Secondary tolerances are 0.1, 0.25 and 1.0. Forty paired randomizations of pool selection and query order are used for each classifier case and ceiling. Every comparator shares the seeds, participants and windows. Complete labels define the evaluation reference; an oracle-based replay checks that acquisition decisions use only the labels already queried.

Test inputs supply the monitoring scores, with true labels hidden during acquisition. The stopping statistic and Window extra metric average over monitoring windows; reported coverage and set size give equal weight to participant summaries. Reference-label savings exclude training, segmentation and setup costs.

4.4. Metrics, uncertainty and ablations

Primary outcomes are queried-label savings $1 - T/N$, reference-set containment failure and batch size inflation. A containment failure is the conservative scalar event $\bar{U}_T < q_{\text{ref}}$; with ties or gaps in candidate scores, this can flag a threshold failure even when no observed test set loses a label. Bracket failure checks both endpoints. Size failure checks whether actual mean extra labels on the monitoring batch exceed τ . Each query campaign is counted once, regardless of the number of held-out people.

For a test participant with empirical coverage c_u , positive shortfall is $(0.9 - c_u)_+$. The positive part is calculated before averaging randomizations. Accuracy and macro-F1 describe the frozen point classifiers. Macro-F1 includes all activity classes, with zero contribution for undefined class scores. Subject summaries average paired randomizations and repeated partitions; paired 95% bootstrap intervals resample these summaries within each dataset. The intervals therefore describe participant variation conditional on the fitted models and partitions. Sex and gender were not used for modeling or stratification, and performance within sex or gender groups was not assessed.

JRC, Bonferroni HG and the confidence sequence based on the prior-posterior ratio use identical tolerances and query checkpoints. Ablations remove the running intersection, use a sparse checkpoint grid, omit repeated-look correction or return the midpoint at the proposed stopping time. The uncorrected and midpoint variants serve as diagnostic controls. Fixed fractions and a one-look HG rule provide additional label-cost and set-size trade-offs. Exploratory checks vary δ and compare monitoring on even-indexed inputs with evaluation on odd-indexed inputs in the first MR case of each development dataset.

All runs use an Intel Core i9-12900HX CPU and an RTX 4070 Laptop GPU with 8 GB memory for neural training. We measure boundary-table construction separately from online acquisition calculations. Human annotation duration and device energy were outside the computational benchmark. Exhaustive small-population enumeration checks both rank engines and the coupled recursion; additional checks cover ties, census behavior, oracle access and strictly monotone score transformations.

4.5. Additional HHAR confirmation

After the main seven-dataset study, an additional Heterogeneity Human Activity Recognition (HHAR) confirmation [37] was specified and publicly committed on 14 September 2026 before preprocessing outcomes or model fitting. The frozen method and classifier implementations were retained. All smartphone-accelerometer streams were processed separately by user and device. Five-second nonoverlapping windows were formed within activity-consistent runs, splitting at non-increasing timestamps or gaps above 0.5 s. Windows required at least 60 finite samples and 4.9 s of observed duration. Timestamp interpolation at 200 Hz followed by anti-aliased resampling produced 128 points. A label-blind random cap of 1,200 windows per user retained 10,800 windows from nine users and six classes. The fixed continuity filter retained very few Nexus4 windows. Device-specific retention counts are provided in Supplementary material 1.

Five declared seeds and three user-disjoint test folds used four training, two calibration and three test users per fold, yielding 45 additional LR, MR and IT cases. The primary analysis used LAC, $N = 240$, $\delta = 0.05$, $\tau = 0.5$ and 40 paired query orders; $N = 120$ and APS were secondary. Alongside Bonferroni HG, the fixed Beta(9, 1) confidence-sequence prior was a primary comparator, chosen from the preceding diagnostic before HHAR outcomes. User-level paired differences and 95% bootstrap intervals follow the declared aggregation. This cohort is reported separately from the main USC-HAD confirmation.

5. Results

5.1. Label savings against sequential quantile baselines

Table 3 reports absolute reference utility and label savings for the three interval methods. Figure 2 adds the secondary APS score. All three methods use the same nominal confidence error, tolerance, pool and query order. Their boundary constructions determine the observed acquisition costs. Figure 2 averages the declared pool ceilings and participant summaries; paired uncertainty is reported in Section 5.3.

On USC-HAD, the joint engine saves 23.67% of reference labels, compared with 17.91% for HG and 11.74% for the uniform-prior CS. Its mean coverage is 91.80%, against 90.36% for the reference, and mean set size changes from 2.590 to 2.769. The coverage increase comes from the more inclusive calibration threshold; the point classifiers are unchanged.

Development-dataset savings range from 4.19% on DSADS to 71.62% on HAR; MHEALTH yields 45.17% and WISDM 6.59%. These outcomes can be read through the precision certificate: querying resolves candidate labels whose scores lie between the current endpoints. Reference size

Table 3: Primary LAC results at confidence error 0.05 and tolerance 0.5

Dataset	Ref. cov. (%)	Ref. size	Joint saved (%)	HG saved (%)	CS saved (%)	Joint extra size
UCI HAR	90.45	1.035	71.62	64.93	58.97	0.235
DSADS	88.69	5.015	4.19	2.71	1.10	0.073
MHEALTH	89.07	1.459	45.17	38.25	30.92	0.206
WISDM	89.92	8.996	6.59	4.35	2.49	0.122
PAMAP2	87.14	2.933	16.20	12.02	6.77	0.151
REALDISP	89.80	2.377	18.56	13.77	8.82	0.163
USC-HAD	90.36	2.590	23.67	17.91	11.74	0.179

Saved (%) is $100(1 - T/N)$ relative to the full reference pool. Coverage, set size and extra size use equally weighted participant summaries; extra size is relative to the full reference. HG: Bonferroni hypergeometric; CS: prior-posterior rank confidence sequence.

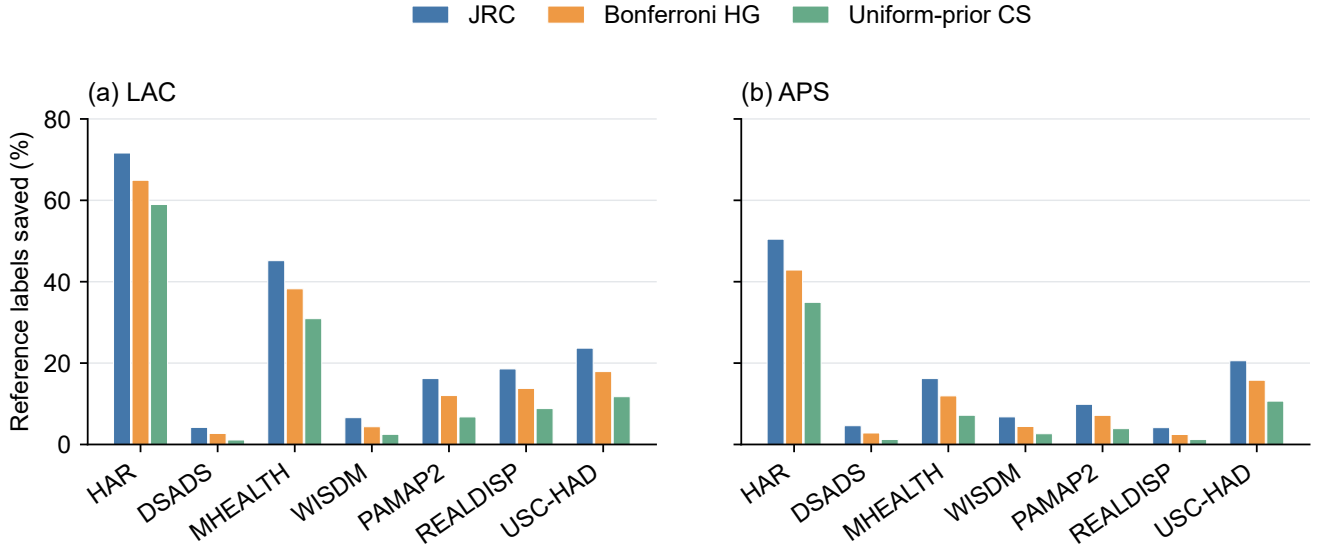


Figure 2: Reference-label savings at $\delta = 0.05$ and $\tau = 0.5$: (a) LAC; (b) APS. Bars show JRC, Bonferroni HG and the uniform-prior CS from left to right

describes the sets being preserved, while this local candidate-score mass determines whether the current bracket meets the common tolerance.

5.2. Reference preservation and absolute utility

Table 4 and Fig. 3 compare fixed-fraction reductions with the quantile-interval baselines. Fixed fractions provide substantial label savings by construction, but their ordinary prefix quantiles need not contain the full-reference sets. A midpoint at a certified stopping time similarly discards the reason that upper-endpoint containment holds. The one-look HG rule returns the upper endpoint of a two-sided interval at the predetermined 50% query fraction; it does not impose the adaptive set-inflation tolerance.

The proposed confirmation threshold-failure rate is 2.04%, and its batch size-failure rate is 0.72%. The confidence statement applies to one query campaign. A failure of the scalar threshold bracket may not remove any label on the finite observed input batch if no candidate score lies in the gap.

Reference quality and acquisition efficiency are connected. Coverage and set size in Tables 3 and 4 describe the predictor being preserved, while the savings and failure columns describe the additional acquisition procedure.

Table 4: USC-HAD primary results and run-level failures

Policy	Saved (%)	Coverage (%)	Set size	Window extra	Threshold fail (%)	Size fail (%)
Full reference	0.00	90.36	2.590	0.0000	0.00	0.00
Joint rank engine	23.67	91.80	2.769	0.1824	2.04	0.72
Bonferroni HG	17.91	91.86	2.774	0.1870	0.45	0.21
Uniform-prior CS	11.74	91.76	2.761	0.1735	0.02	0.05
Fixed 25%	75.00	90.46	2.713	0.1220	42.26	15.03
Fixed 50%	50.00	90.41	2.632	0.0416	41.42	6.09
Fixed 75%	25.00	90.36	2.601	0.0102	39.83	2.20
One-look HG (50% look)	50.00	93.83	3.356	0.7737	1.73	46.64
Midpoint output	23.67	90.04	2.562	-0.0282	46.40	0.14
No time correction	33.71	91.73	2.766	0.1784	6.29	3.02

Saved (%) is $100(1 - T/N)$ relative to the full reference pool. Coverage and set size use participant weighting; Window extra is mean added set size over monitoring windows. Threshold fail: $\bar{U}_T < q_{\text{ref}}$; Size fail: window inflation > 0.5 . Failure rates average query campaigns. One-look HG uses 2.5% per tail and returns the upper endpoint.

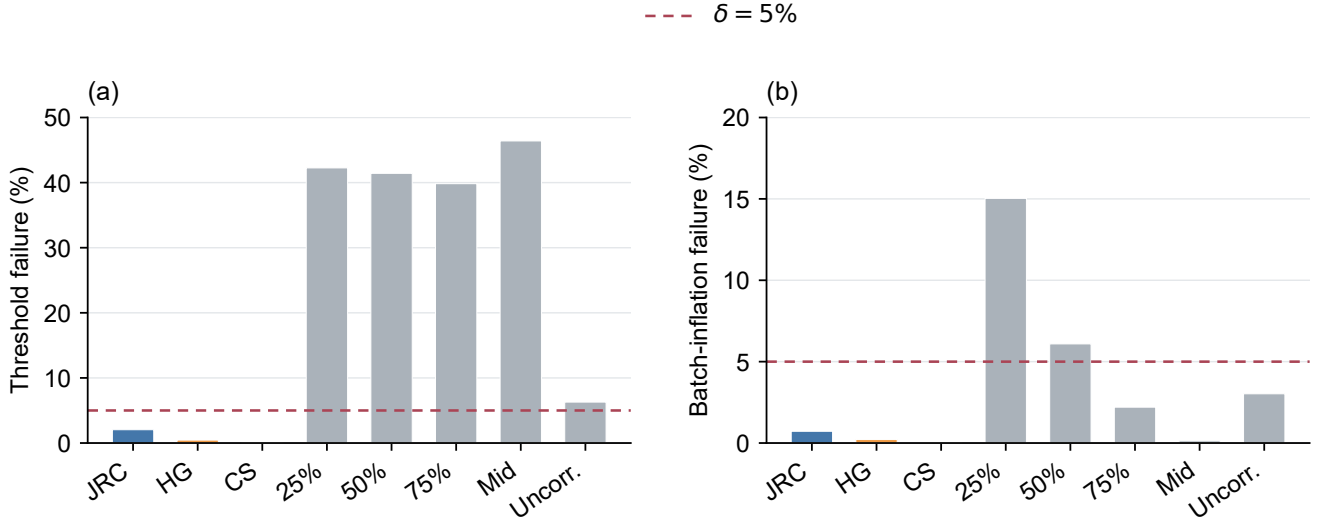


Figure 3: USC-HAD LAC failures: (a) threshold failure; (b) monitoring-batch inflation above 0.5 labels. Dashed line: $\delta = 5\%$. Gray bars denote diagnostic rules; percentages denote query fractions. Mid: midpoint output; Uncorr.: uncorrected repeated looks

5.3. Paired improvements and classifier dependence

Figure 4 reports paired participant bootstrap intervals for additional savings against the quantile-interval baselines. These intervals condition on the fitted models and split collection. Table 5 separates point-classification performance from acquisition outcomes for each confirmation backbone. Acquisition uses their frozen predictions.

Table 5: USC-HAD point-classifier and acquisition results

Classifier	Accuracy (%)	Macro-F1 (%)	Labels saved (%)	Set coverage (%)	Set size
LR	60.17	56.14	32.85	92.76	2.769
MR	68.04	65.40	23.31	92.82	2.573
IT	65.14	60.32	14.85	89.83	2.967

Each classifier has 35 outer partitions. IT is one InceptionTime-style network; macro-F1 includes all 12 classes.

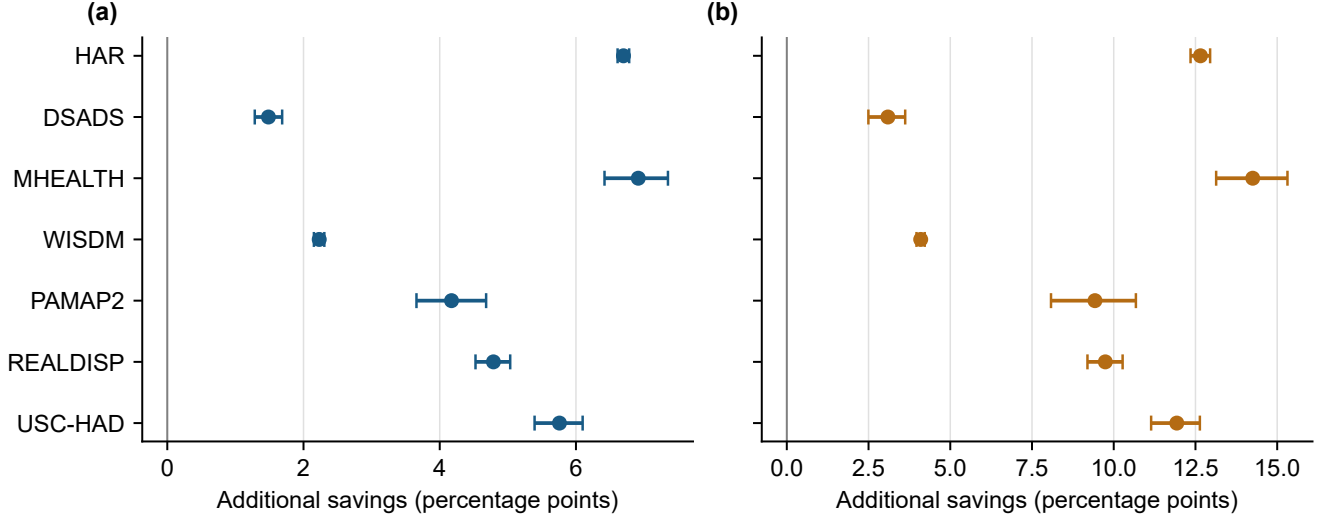


Figure 4: Additional LAC label savings of JRC over (a) Bonferroni HG and (b) the uniform-prior CS. Points show paired participant means; bars show 95% bootstrap intervals

5.4. Tolerance, query checkpoints and mechanism ablations

Figure 5 reports mean savings at all four tolerances declared before the USC-HAD results. Paired uncertainty at the primary tolerance is shown in Fig. 4. Increasing the tolerance enlarges the set of brackets that certify sufficient precision in Proposition 2 and therefore allows earlier stopping along the same query path. The primary tolerance is 0.5.

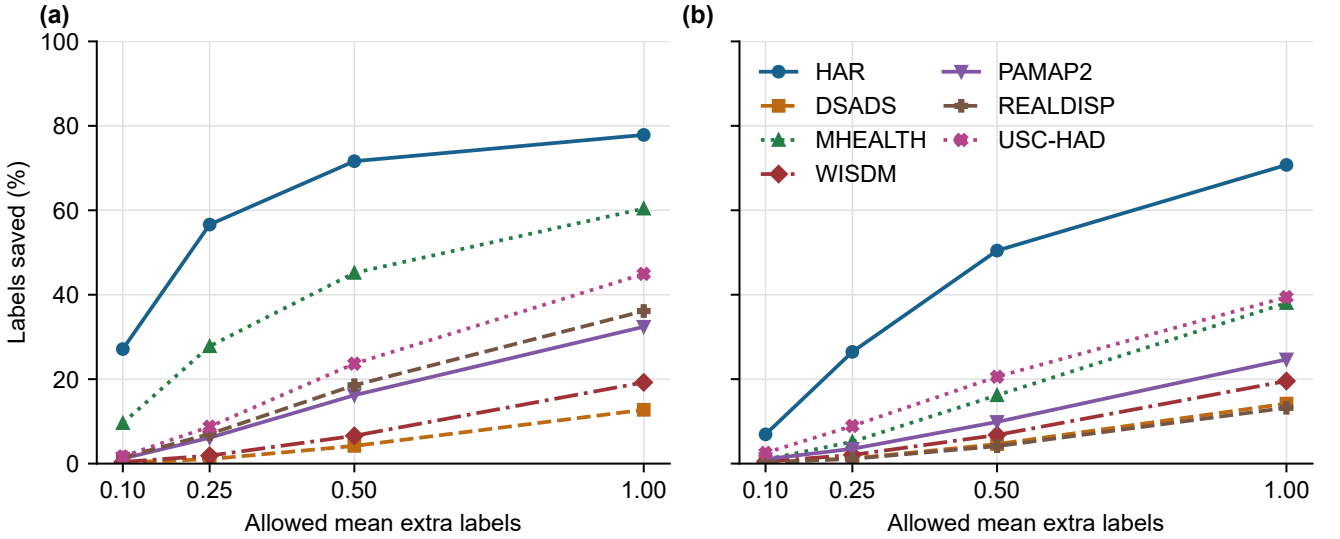


Figure 5: Mean label savings of JRC versus the allowed extra set size: (a) LAC; (b) APS. Markers and line styles identify datasets

Table 6 separates the mechanism choices. The first three rows retain their stated boundaries and checkpoints; midpoint and uncorrected variants are diagnostics. Removing running intersections reduces confirmation savings from 23.67% to 21.10%; sparse checkpoints reduce them to 16.84%. Returning the midpoint retains the same label count but raises threshold failure to 46.40%, illustrating why the protected upper endpoint is essential. Omitting repeated-look correction increases savings to 33.71% and threshold failure to 6.29%, above the declared 5% error level.

Table 6: USC-HAD mechanism ablations

Rule	Saved (%)	Extra size	Threshold fail (%)	Size fail (%)
Joint rank engine	23.67	0.179	2.04	0.72
No intersection	21.10	0.178	0.67	0.06
Sparse checkpoints	16.84	0.079	1.77	0.36
Midpoint output	23.67	-0.028	46.40	0.14
No time correction	33.71	0.175	6.29	3.02

Saved (%) is $100(1 - T/N)$ relative to the full reference pool. Extra size uses participant weighting and is relative to the reference. The first three rows retain their stated guarantees; midpoint and uncorrected rows are diagnostics.

Figure 6 traces the first declared USC-HAD MR query campaign at a 480-window pool. The trace uses the fixed first case. The census collapses all valid constructions to the same full-reference quantile, approximately 0.999436 in this example. The plotting floor is 10^{-16} . Before that point, the mapping from rank uncertainty to set membership determines when each engine meets the tolerance.

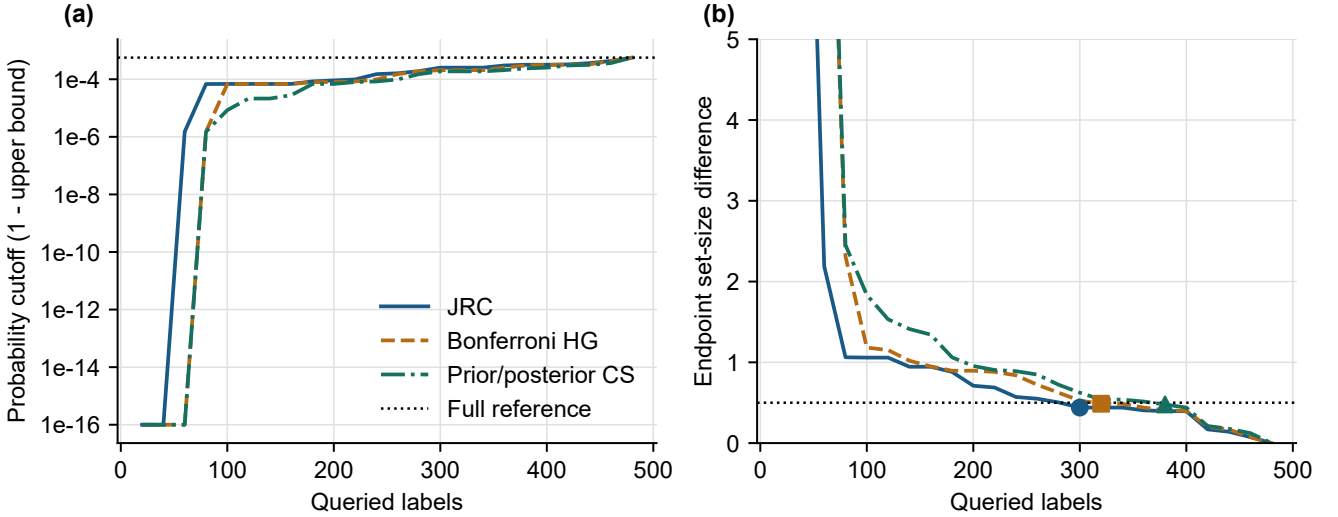


Figure 6: USC-HAD MR query trace for LAC, $N = 480$: (a) probability cutoff $1 - \bar{U}_t$ on a logarithmic scale; (b) endpoint set-size difference, with stopping locations marked. Zero cutoffs and infinite upper bounds are plotted at 10^{-16} . Horizontal line: $\tau = 0.5$

Figure 7 relates label savings to absolute reference set size and shows the corresponding reference and returned-set coverage pairs. Each dataset contributes LR, MR and IT points; their identities and values are provided in the model-summary table of Supplementary material 1.

5.5. Sensitivity to stronger confidence-sequence priors

The primary confidence-sequence baseline uses a uniform prior on the finite-population success count. After the main confirmation, we evaluated four fixed beta-binomial priors across all 354 existing classifier cases. A $\text{Beta}(a, b)$ mixture replaces the ordered-prefix mixture likelihood by $B(j+a, t-j+b)/B(a, b)$ while retaining the same likelihood-ratio construction and error allocation. Exact rank-path checks verify the error bound for each finite design used. The diagnostic reuses the main models and queries after confirmation; the locked JRC rule is unchanged.

Table 7 reports all four fixed prior choices. On USC-HAD, the strongest of these variants saves 15.33% of labels, compared with 11.74% for the uniform prior and 23.67% for JRC. The variation demonstrates the role of boundary construction in acquisition efficiency under the declared set-

tings. Reconstructing JRC and the uniform-prior results reproduces the archived dataset averages within 10^{-12} .

5.6. Schema exclusion and computation

The exclusion sensitivity removes the USC-HAD trial with conflicting activity identifiers from every data role. The original classifier is reused only when that trial was absent from its training set; otherwise it is refitted. Joint-engine label savings change from 23.67% to 22.70%, coverage from 91.80% to 92.30%, and mean set size from 2.769 to 2.814. This check quantifies sensitivity to the trial-level label discrepancy.

Boundary tables depend only on the finite design and can be reused. Table 8 separates table construction from execution with a cached table. Timings exclude JIT compilation, model inference and human labeling. Cached query execution includes sorting monitoring scores and observed prefixes. The timing benchmark uses a fixed score array and 1,000 unlabeled inputs with 12 candidate classes.

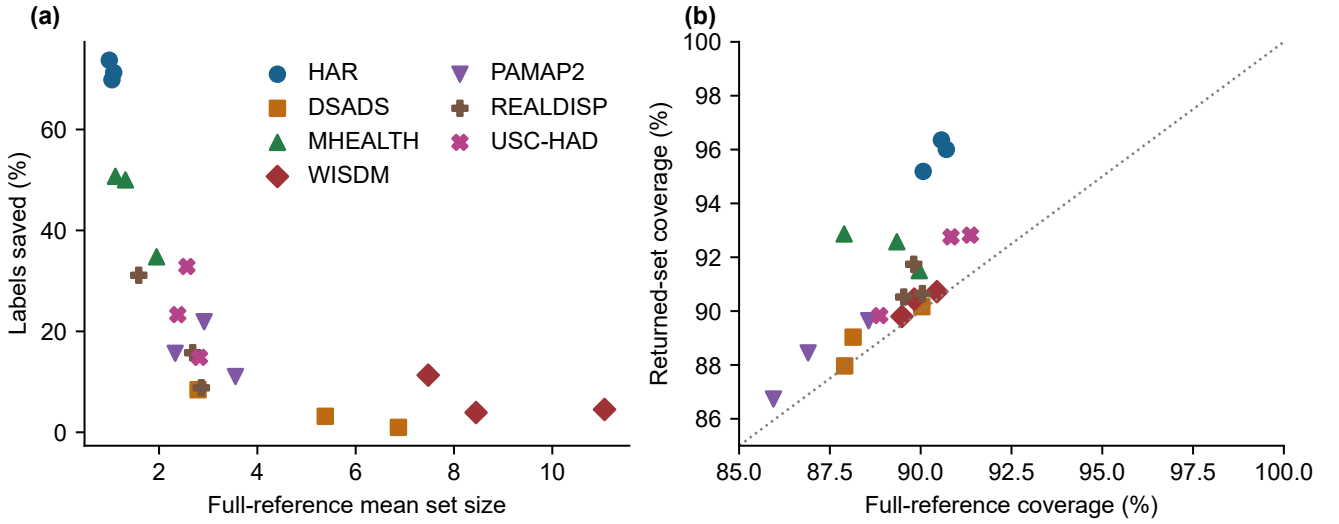


Figure 7: Reference utility for LAC: (a) label savings versus reference set size; (b) returned-set versus reference coverage. Each point is a dataset-classifier mean; color and shape identify datasets. The diagonal denotes equal coverage

Table 7: Post-confirmation prior sensitivity: percentage of reference labels saved

Dataset	JRC	CS(1, 1)	CS(0.5, 0.5)	CS(1.8, 0.2)	CS(9, 1)
UCI HAR	71.62	58.97	60.85	66.89	64.84
DSADS	4.19	1.10	1.12	2.09	1.82
MHEALTH	45.17	30.92	32.06	36.58	36.51
WISDM	6.59	2.49	2.53	3.02	3.34
PAMAP2	16.20	6.77	6.95	9.24	9.20
REALDISP	18.56	8.82	9.16	10.86	11.73
USC-HAD	23.67	11.74	12.01	14.22	15.33

LAC at $\delta = 0.05$ and $\tau = 0.5$. $CS(a, b)$ uses a $\text{Beta}(a, b)$ mixture; values use participant weighting. This diagnostic follows confirmation.

Table 8: Boundary construction and cached acquisition computation

Pool size	Build table (ms)	Cached query loop (ms)	Exact crossing probability
120	8.31	0.18	0.04843
240	25.07	0.26	0.04606
480	53.80	0.50	0.04991
960	153.41	0.45	0.04981

Five repetitions on the stated CPU at $\delta = 0.05$. Build time is per pool/checkpoint design; query time uses cached boundaries.

5.7. Additional HHAR confirmation

Table 9 reports HHAR label savings of 34.03% for JRC, 28.58% for HG and 25.13% for the pre-specified Beta(9, 1) CS. Paired gains are 5.45 percentage points (95% CI 5.09–5.85) against HG and 8.89 (8.52–9.23) against CS. These participant-bootstrap intervals condition on the fitted models and partitions.

Reference coverage and mean set size are 88.10% and 1.807, versus 90.84% and 1.997 for JRC. Table 9 includes both failure rates; secondary pool-size, APS and classifier results are in Supplementary material 1.

Table 9: Additional HHAR primary confirmation, $N = 240$

Policy	Saved (%)	Threshold fail (%)	Inflation fail (%)
Reference	0.00	0.00	0.00
JRC	34.03	1.33	0.28
HG	28.58	0.39	0.00
CS(9, 1)	25.13	0.28	0.00
CS(1, 1)	20.64	0.06	0.00

Saved (%) is $100(1 - T/N)$ relative to the full reference pool. CS(a, b) denotes the Beta prior; failure rates count query campaigns.

6. Discussion

Reference preservation connects calibration-label acquisition to the precision of the resulting decisions. The endpoint gap counts candidate labels whose membership is unresolved by the current quantile interval. Proposition 2 identifies it as the smallest worst-case mean enlargement under that interval’s containment requirement. This explains the operational role of the lower endpoint: together with the upper endpoint it determines whether the stated precision can be certified.

The statistical and decision components play complementary roles. Checkpoint calibration controls the probability that the quantile leaves the bracket, while the endpoint gap measures the consequences of the remaining interval uncertainty. The experiments vary the boundary construction, query schedule and tolerance within a common acquisition protocol. Their reference-size, coverage and query-count results characterize the cost of preserving different set-valued predictors.

The procedure conditions on a fixed reference pool, uniform queries and an available monitoring batch. These conditions make the rank law explicit and the precision certificate observable. Further work can incorporate grouped acquisition, changing monitoring batches and measured

annotation effort. The current computational study reports queried-label counts, with participant setup represented by the stated cost model.

7. Conclusion

Sequential reference preservation makes prediction-set precision a criterion for calibration-label acquisition. The endpoint set-size gap provides a tight enlargement certificate under interval uncertainty, and joint rank calibration supplies bounds over the declared query checkpoints. Their combination preserves the complete-reference sets and controls average additional labels on the monitoring batch on a common confidence event. The wearable evaluations quantify the resulting label savings and their dependence on reference quality, boundary construction and tolerance.

CRedit authorship contribution statement

Xin Feng: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Writing – original draft. **Decong Yang:** Visualization, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics and consent

This study analyzes existing public benchmark data and involved no new participant recruitment or intervention.

Data availability

The eight source datasets are available from the cited providers [28–34, 37]. Frozen model predictions, query-campaign records, evaluation tables and protocols are publicly available in the versioned reproducibility archive [38]. Supplementary material 1 contains the same scientific records together with verification results and reproduction instructions.

Code availability

Code and reproduction instructions are available at <https://github.com/Cyrano666/jrc-reference-emulation>. The reported experiments use the archived implementation accompanying the frozen numerical records [38].

Supplementary material

Supplementary material 1: reproduction code, frozen predictions, evaluation tables, protocols, verification records and editable figure sources.

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